

Role-Occupancy Closure under Admissibility

Why role-cardinality matching is only the cardinality shadow of compatible realization

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Abstract

This paper introduces a global constraint in the Admissibility And Standing Constraint (AASC) framework governing the transition from admissible candidate residue to realized target occupation. The paper does not generate a new closure mechanism. It tightens the residue constraint by exhausting the remaining ways a fixed certificate can satisfy candidate admissibility while failing to realize the forced target role architecture. Singleton-residue closure is the one-role special case; bare role-cardinality matching is still too weak. Equal numbers of candidates and roles do not license realization if the candidates are eligible for the wrong roles, if they are jointly incompatible, or if their apparent uniqueness is produced only by an inadmissible relabeling.

The resulting closure condition is Role-Occupancy Closure. For a fixed certificate C , target T , forced role set $\mathcal{R}_T(C, T)$, presentation-quotiented admissible residue $\text{Residue}(C)$, same-certificate eligible target-relative residue

$$\text{Residue}_T^{\text{elig}}(C, T) := \{x \in \text{Residue}(C) : \exists r \in \mathcal{R}_T(C, T) E_C(x, r)\},$$

and live occupation-capable target-relative residue $\text{Residue}_T^{\text{live}}(C, T)$, the compatible assignment set is

$$\Sigma_C(T) = \left\{ \sigma : \mathcal{R}_T(C, T) \xrightarrow{\sim} \text{Residue}_T^{\text{live}}(C, T) : \forall r \in \mathcal{R}_T(C, T) E_C(\sigma(r), r) \text{ and } J_C(\sigma) = 1 \right\}.$$

The certificate-lawful assignment invariant tuple $\mathfrak{I}_{C, T}$ records the certificate, target, live residue, eligibility, compatibility, standing, tensor, boundary, continuation, failure, and lawful-redescription data; the automorphism group is $\Gamma_{C, T}^{\text{lawful}} = \text{Aut}(\mathfrak{I}_{C, T})$. Thus the quotient removes only transformations preserving that invariant tuple. The central theorem is

$$\text{RealizationLicensed}(C, T) \iff |\Sigma_C(T)/\Gamma_{C, T}^{\text{lawful}}| = 1.$$

Cardinality equality is therefore a necessary consequence of closure, not the closure theorem itself. The paper proves necessity and sufficiency separately, gives formation rules for E_C , J_C , live residue, and certificate-lawful automorphisms, proves a fixed-certificate constraint-diagnosis exhaustion theorem plus explicit scope reset, and uses the result to diagnose Dirac-style paired solutions, neutrino mass ordering, charged-lepton pole transport, cosmological parameter transport, and Yang–Mills sector extension. All uniqueness and non-existence claims are restricted to the fixed certificate C , target T , and governance-role equivalence on that domain.

Notation guide

Symbol	Working meaning
C	fixed certificate: the declared admissibility environment
T	target role architecture under evaluation
$\mathcal{R}_T(C, T)$	forced roles that must be occupied for target realization
$R_{\text{adm}}(C)$	candidates surviving the declared admissibility tests
$\sim_{\text{pres}}$	presentation-equivalence: skin, notation, coordinate, gauge, phase, and bookkeeping collapse only
$\text{Residue}(C)$	presentation-quotiented admissible residue
$\text{Residue}_T^{\text{elig}}(C, T)$	target-relative residue with at least one same-certificate eligibility edge
$\text{Residue}_T^{\text{live}}(C, T)$	occupation-capable residue occurring in some jointly compatible complete pre-assignment
$E_C(x, r)$	residue x is same-certificate eligible for role r
$J_C(\sigma)$	full assignment σ is jointly compatible under tensor, boundary, continuation, and failure constraints
$\Sigma_C(T)$	certificate-compatible bijections from forced roles onto live target-relative residue
$\Gamma_{C,T}^{\text{lawful}}$	certificate-lawful role-incidence automorphisms preserving the invariant tuple $\mathcal{I}_{C,T}$

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1 Introduction: constraint tightening by role occupancy

This paper does not introduce a generator of realization. It tightens the existing residue constraint under fixed-domain admissibility by exhausting the remaining ways a certificate can satisfy candidate admissibility while failing to realize the forced target role architecture. The operative condition, Role-Occupancy Closure, is the residue that remains after weaker formulations collapse under the kernel invariants: candidate production is not standing production; branch separation is not branch selection; cardinality equality is not role realization; and empirical success is not a same-scope selector.

A certificate may generate candidate responses, separate them, display their differences, and diagnose their degeneracies while still failing to realize any candidate as an occupant of the target role. The singleton-residue condition is recovered only as the one-role specialization:

no realization without singleton residue.

This condition is appropriate for one-role targets, but it is not a global realization theorem. Some targets require two complementary occupants, three family occupants, a k -multiplet, or a sectoral layer over a local theorem object. In such cases, multiple surviving residues are not automatically excess alternatives. They may be exactly the residue required by the target role architecture. The paper’s contribution is therefore eliminative: it identifies the exact constraint remaining once singleton residue, bare cardinality matching, pointwise eligibility, and unconstrained symmetry quotienting have each been shown insufficient.

The Dirac-style warning is useful here, but only schematically. A doubled formal solution structure should not be classified in advance as failure merely because it is not singleton. The relevant question is not whether one formal solution must be eliminated. The relevant question is whether the target role architecture is one-role, paired-role, family-role, multiplet-role, or sector-role. The present manuscript does not derive the positron and does not use Dirac theory as evidence. It uses the example only to block singleton absolutism.

The general point is this: even cardinality equality is not enough. If a target requires roles A and B , and two surviving residues are both eligible only for A , then cardinality equality holds but the target role structure is not realized. If one residue is eligible for A and the other for B , but the pair violates a tensor or continuation condition, cardinality equality still does not license realization. Realization therefore requires unique compatible role occupancy, not bare residue cardinality.

1.1 The general thesis

The general thesis is a constraint-tightening theorem, not a constructive mechanism. AASC supplies no generator for role occupancy. It constrains the admissible residue until the forced role set can be occupied in exactly one way modulo certificate-lawful automorphism.

Let C be a fixed certificate. Let $R_{\text{adm}}(C)$ be the admissible candidate response set. Let $\sim_{\text{pres}}$ collapse only presentation-level duplicates, and define

$$\text{Residue}(C) = R_{\text{adm}}(C) / \sim_{\text{pres}} .$$

Let $\mathcal{R}_T(C, T)$ be the forced target role set. Let

$$E_C \subseteq \text{Residue}(C) \times \mathcal{R}_T(C, T)$$

be the same-certificate role-eligibility relation, and let $J_C(\sigma) \in \{0, 1\}$ state whether a complete role assignment is jointly compatible under the tensor, continuation, boundary, and failure conditions of

the certificate. The eligible target-relative residue is

$$\text{Residue}_T^{\text{elig}}(C, T) := \{x \in \text{Residue}(C) : \exists r \in \mathcal{R}_T(C, T) E_C(x, r)\}.$$

A second refinement is needed. Some pointwise eligible residues cannot occur in any jointly compatible complete assignment. Those residues are not live occupants; they are compatibility-impossible and fail closed. The live target-relative residue $\text{Residue}_T^{\text{live}}(C, T)$ is the union of residues appearing in some injective, role-covering, pointwise eligible assignment with $J_C = 1$.

The closure object is the set $\Sigma_C(T)$ of compatible bijections from the forced role set onto $\text{Residue}_T^{\text{live}}(C, T)$. Realization is licensed iff there is exactly one such assignment up to certificate-lawful role-incidence automorphism:

$$\text{RealizationLicensed}(C, T) \iff \left| \Sigma_C(T) / \Gamma_{C, T}^{\text{lawful}} \right| = 1.$$

Thus role-cardinality matching is the cardinality shadow of a stronger theorem: unique compatible bijective role occupancy over the live target-relative residue.

1.2 Core nonclaims

Nonclaim 1.1 (No universal singleton claim). The paper does not claim that every realized domain requires singleton residue. Singleton residue is the special case in which $|\mathcal{R}_T(C, T)| = 1$.

Nonclaim 1.2 (No bare cardinality sufficiency). The paper does not claim that $|\text{Residue}_T^{\text{live}}(C, T)| = n_{\text{role}}(C, T)$ alone licenses realization. Equal live target-relative cardinality is necessary for finite closure, but it is not sufficient without role eligibility, joint compatibility, and uniqueness modulo certificate-lawful incidence automorphism.

Nonclaim 1.3 (No generator for role occupancy). AASC supplies no generator for role occupancy. It constrains the admissible residue until the forced role set can be occupied in exactly one way modulo certificate-lawful automorphism. Any appearance of selection or generation is diagnosed as degeneracy: a selector gap, incidence failure, compatibility failure, impossible residue, or explicit scope reset.

Nonclaim 1.4 (No empirical tie-break). The theorem does not license empirical fit, likelihood ranking, posterior success, semantic intention, explanatory elegance, or convenience as a same-scope selector. Empirical material may enter only when declared as part of the certificate, for example as a stable bivalent boundary trace or record.

2 From singleton residue to role occupancy

2.1 The singleton formulation

The singleton criterion can be stated as

$$\text{RealizationLicensed}(C, T) \Rightarrow |\text{Residue}(C)| = 1.$$

A refined form first quotients redundant presentations. That form is valid when the target architecture has exactly one role occupant: a single mass-ordering branch in one realized universe, a single protected pole-trace remainder, or a single same-target transport bridge.

2.2 Why singleton residue is target-restricted

If the target demands two complementary roles, three family roles, a multiplet with k required occupants, or a sector layer with several independently typed objects, singleton residue misclassifies legitimate role multiplicity as failure. The finite cardinality shadow of closure is

$$|\text{Residue}_T^{\text{live}}(C, T)| = |\mathcal{R}_T(C, T)| = n_{\text{role}}(C, T),$$

but this is not yet the theorem. The theorem requires live target-relative residue coverage, pointwise eligibility, joint compatibility, and uniqueness modulo certificate-lawful role-incidence automorphism.

2.3 Why cardinality-only is still too weak

Suppose

$$\mathcal{R}_T(C, T) = \{A, B\}, \quad \text{Residue}_T^{\text{live}}(C, T) = \{x, y\},$$

and both x and y are eligible only for role A . Then cardinality equality holds, but role B is unpopulated. Conversely, suppose x is eligible for A and y for B , but the pair (x, y) violates a tensor or continuation compatibility constraint. Again cardinality equality holds, but realization fails. Therefore the main object is not a cardinality equation, but a unique compatible role-covering assignment.

Remark 2.1 (The answer is not chosen). The answer is not chosen from a candidate list. The answer is the unique certificate-compatible occupation of the forced target role set by the live target-relative residue. If the target needs one occupant and two eligible rivals survive, the result is underdetermined. If the target needs two complementary occupants and there is a unique jointly compatible assignment modulo role-incidence symmetry, the result is closed.

3 Fixed-domain role architecture

All definitions are certificate-relative unless explicitly stated otherwise.

Definition 3.1 (Fixed certificate). A fixed certificate is the declared admissibility-bearing comparison environment

$$C = (D, \text{Roles}, \text{Adm}, \sim, \text{Boundary}, \text{Cont}, \text{Skin}, \text{Failure}, \text{Support}).$$

It fixes the declared domain, target role space, admissible transformations, candidate-response construction, boundary records, continuation rules, skin/display status, failure conditions, and permitted support material.

A result is relative to C . Changing the domain, role vocabulary, admissibility predicate, boundary record, support, quotient rule, or continuation structure is not same-scope repair. It is explicit scope reset.

Definition 3.2 (Candidate response class). A candidate response class is a possible occupant, branch, transport, correction, sector, readout, or response generated by C . The admissible candidate set is

$$R_{\text{adm}}(C) = \{R_i : R_i \text{ satisfies the declared admissibility tests under } C\}.$$

Candidate status is not realized status.

Definition 3.3 (Presentation equivalence). Two admissible candidates are presentation-equivalent, written

$$R_i \sim_{\text{pres}} R_j,$$

when they differ only by skin, notation, phase convention, gauge presentation, coordinate choice, redundant bookkeeping label, or another admissible redescription that changes no role eligibility, no standing-bearing continuation, and no quotient-level tensor content.

Definition 3.4 (Admissible residue). The admissible residue of C is

$$\text{Residue}(C) = R_{\text{adm}}(C) / \sim_{\text{pres}}.$$

It is the set of presentation-quotiented surviving candidates. It is not yet a role assignment.

The move from role-equivalence to presentation-equivalence is deliberate. A quotient that already says two candidates “occupy the same target role” risks assuming the role assignment that the theorem is meant to test. The residue quotient therefore collapses only presentation-level duplication. Role occupancy is handled separately by role eligibility and assignment.

Definition 3.5 (Forced target role set). The forced target role set is

$$\mathcal{R}_T(C, T) = \{r : r \text{ is a role forced by the target architecture } T \text{ under } C\}.$$

The target role-cardinality is

$$n_{\text{role}}(C, T) := |\mathcal{R}_T(C, T)|.$$

The set $\mathcal{R}_T(C, T)$ must be fixed before residue matching. It cannot be chosen after inspecting $\text{Residue}(C)$.

Definition 3.6 (Role eligibility relation). The role-eligibility relation is

$$E_C \subseteq \text{Residue}(C) \times \mathcal{R}_T(C, T).$$

The statement $E_C(x, r)$ means that residue class x is admissible as an occupant of target role r under certificate C .

Definition 3.7 (Formation rules for E_C). A positive eligibility edge $E_C(x, r)$ may be introduced only by one of the following certificate-typed sources:

- (i) internal role typing fixed by the target architecture;
- (ii) provenance or indexical typing declared before role use;
- (iii) a declared stable bivalent boundary trace or role trace.

A scope reset does not form an edge of E_C under fixed C . It produces a new certificate C' and a new relation $E_{C'}$, after which the theorem must be reapplied to the new pair.

The following do not form eligibility edges: likelihood preference, posterior success, explanatory elegance, scalar ranking, convention, semantic intention, institutional preference, downstream utility, or post-hoc repair.

Definition 3.8 (Eligible target-relative residue). The eligible target-relative residue is

$$\text{Residue}_T^{\text{elig}}(C, T) := \{x \in \text{Residue}(C) : \exists r \in \mathcal{R}_T(C, T) E_C(x, r)\}.$$

Only residues with at least one same-certificate eligibility edge for the target enter this set. A residue generated by the broad certificate but ineligible for every role of T is outside the target-relative assignment problem.

Definition 3.9 (Joint compatibility predicate). For a candidate role assignment $\sigma : \mathcal{R}_T(C, T) \rightarrow \text{Residue}_T^{\text{elig}}(C, T)$, the value

$$J_C(\sigma) \in \{0, 1\}$$

states whether the full assignment is jointly admissible under the certificate. The predicate includes tensor compatibility, continuation compatibility, boundary-record compatibility, failure-condition satisfaction, and role co-instantiation constraints. It is not reducible to pointwise eligibility unless the certificate proves such factorization:

$$J_C(\sigma) \not\equiv \prod_{r \in \mathcal{R}_T(C, T)} E_C(\sigma(r), r)$$

by default. Two pointwise eligible occupants can be jointly inadmissible.

Definition 3.10 (Formation rules for J_C). The predicate $J_C(\sigma) = 1$ may be certified only by certificate-declared co-instantiation rules, including:

- (i) anchor/tensor/skin tensor compatibility on the fixed carrier;
- (ii) continuation compatibility across the retained construction trajectory;
- (iii) boundary-record and trace compatibility;
- (iv) failure closure and impossibility exclusion;
- (v) explicitly declared co-occupation constraints of the target role architecture.

It may not be supplied by likelihood ranking, success, elegance, posterior comparison, preference ordering, scalar proximity, delayed calibration, or post-hoc repair.

Definition 3.11 (Pre-compatible occupation assignment). A pre-compatible occupation assignment is an injective map

$$\tau : \mathcal{R}_T(C, T) \hookrightarrow \text{Residue}_T^{\text{elig}}(C, T)$$

such that

$$\forall r \in \mathcal{R}_T(C, T) \ E_C(\tau(r), r), \quad J_C(\tau) = 1.$$

Let $\Sigma_C^{\text{pre}}(T)$ denote the set of such assignments. The injectivity condition is part of the definition: distinct forced roles require distinct residues unless the target architecture has already quotiented or retyped the apparent roles before $\mathcal{R}_T(C, T)$ is fixed.

Definition 3.12 (Live target-relative residue). The live target-relative residue is

$$\text{Residue}_T^{\text{live}}(C, T) := \bigcup_{\tau \in \Sigma_C^{\text{pre}}(T)} \text{im}(\tau).$$

Thus $\text{Residue}_T^{\text{live}}(C, T) \subseteq \text{Residue}_T^{\text{elig}}(C, T)$. A residue in

$$\text{Residue}_T^{\text{elig}}(C, T) \setminus \text{Residue}_T^{\text{live}}(C, T)$$

is pointwise eligible for at least one role but cannot occur in any jointly compatible complete occupation assignment. Such a residue is diagnosed as compatibility-impossible under fixed C , not as a live residue left for selection.

Lemma 3.13 (Live-residue union detects assignment alternatives). *A residue is live exactly when it occurs in some injective, pointwise eligible, jointly compatible complete occupation attempt for (C, T) . If a live residue is not used by a proposed realized assignment, then the certificate still contains an alternative occupation attempt, unless the difference is removed by the lawful action of $\Gamma_{C,T}^{\text{lawful}}$. Unused live residue is therefore not inert surplus; it is over-residue or a selector gap.*

Proof. The definition of $\text{Residue}_T^{\text{live}}(C, T)$ is the union of images of assignments in $\Sigma_C^{\text{pre}}(T)$. Membership in that union is therefore equivalent to occurrence in at least one injective, role-covering, pointwise eligible assignment with $J_C = 1$. If such a residue is unused by a proposed realized assignment, the certificate has not merely retained an irrelevant candidate: it has retained a residue that can participate in a complete compatible occupation attempt. If all assignments using that residue lie in the same $\Gamma_{C,T}^{\text{lawful}}$ -orbit as the proposed realized assignment, the difference is lawful relabeling. If not, the difference is an admissibility-relevant alternative assignment and hence a selector gap. Thus the live-residue union intentionally detects assignment-level alternatives rather than merely pointwise eligibility. $\square$

Definition 3.14 (Compatible role-covering assignment). A compatible role-covering assignment is a bijection

$$\sigma : \mathcal{R}_T(C, T) \xrightarrow{\sim} \text{Residue}_T^{\text{live}}(C, T)$$

such that:

- (i) pointwise role eligibility holds: $\forall r \in \mathcal{R}_T(C, T), E_C(\sigma(r), r)$;
- (ii) joint compatibility holds: $J_C(\sigma) = 1$.

Equivalently, it is a role assignment that covers every forced role, uses every live target-relative residue, assigns distinct residues to distinct roles, and satisfies the full joint compatibility predicate. The set of all such assignments is denoted $\Sigma_C(T)$.

Bijection is part of the formal definition. If two apparent roles may lawfully share one occupant, then the target architecture has already quotiented or retyped those apparent roles before $\mathcal{R}_T(C, T)$ is fixed. Shared occupancy is therefore handled by target typing, not by allowing a non-injective assignment inside a fixed role set.

Definition 3.15 (Certificate-lawful assignment invariant tuple). The certificate-lawful assignment invariant tuple for (C, T) is

$$\mathcal{J}_{C,T} := (C, T, \mathcal{R}_T(C, T), \text{Residue}_T^{\text{live}}(C, T), E_C, J_C, \text{St}, \text{Ten}, \text{Bd}, \text{Cont}, \text{Fail}, \sim_{\text{law}}).$$

Here St denotes standing status, Ten tensor content, Bd boundary records, Cont continuation structure, Fail failure conditions, and $\sim_{\text{law}}$ the lawful redescription class fixed by the certificate. The tuple records exactly the certificate and target data whose preservation is required before two assignments may be treated as the same realization up to relabeling.

Definition 3.16 (Certificate-lawful role-incidence automorphism group). The certificate-lawful role-incidence automorphism group is

$$\Gamma_{C,T}^{\text{lawful}} := \text{Aut}(\mathcal{J}_{C,T}).$$

Equivalently, an element is a pair $\gamma = (\alpha, \beta)$, where

$$\alpha : \mathcal{R}_T(C, T) \rightarrow \mathcal{R}_T(C, T), \quad \beta : \text{Residue}_T^{\text{live}}(C, T) \rightarrow \text{Residue}_T^{\text{live}}(C, T)$$

are bijections satisfying all of the following:

- (i) they preserve the full invariant tuple $\mathfrak{I}_{C,T}$, including C , T , $\mathcal{R}_T(C, T)$, and $\text{Residue}_T^{\text{live}}(C, T)$;
- (ii) they preserve role eligibility:

$$E_C(x, r) \iff E_C(\beta x, \alpha r);$$

- (iii) they preserve joint compatibility:

$$J_C(\sigma) = J_C(\beta \circ \sigma \circ \alpha^{-1})$$

for every bijective role assignment σ ;

- (iv) they preserve standing status, tensor content, boundary records, continuation structure, failure conditions, and the lawful redescription class of the certificate.

Thus $\Gamma_{C,T}^{\text{lawful}}$ is not the abstract automorphism group of a bare incidence graph. It quotients only genuine certificate-lawful relabelings preserving $\mathfrak{I}_{C,T}$. It cannot erase physically, constructionally, or tensor-distinct alternatives. It is not a symmetry generator; it is the maximal lawful equivalence that still preserves the admissibility-relevant invariant bundle. Any larger quotient would erase standing-bearing distinctions and collapse to degeneracy.

The group $\Gamma_{C,T}^{\text{lawful}}$ acts on assignments by

$$(\gamma \cdot \sigma)(r) := \beta(\sigma(\alpha^{-1}r)), \quad \gamma = (\alpha, \beta).$$

If both the forced roles and the live target-relative residues are rigidly individuated, $\Gamma_{C,T}^{\text{lawful}} = \{1\}$ and assignment uniqueness is literal. If roles or residues are interchangeable only by certificate-lawful transformations, uniqueness is assessed modulo $\Gamma_{C,T}^{\text{lawful}}$.

Definition 3.17 (Role-Occupancy Closure). Role-Occupancy Closure holds for (C, T) , written $\text{ROC}(C, T)$, iff

$$\left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right| = 1.$$

Equivalently, there is exactly one compatible bijective role-covering assignment up to certificate-lawful automorphism of the role-incidence structure.

Definition 3.18 (Licensed realization). $\text{RealizationLicensed}(C, T)$ holds iff certificate C emits a standing-positive target object for T such that:

- (i) every forced target role in $\mathcal{R}_T(C, T)$ is occupied;
- (ii) every occupant is a target-relative admissible residue;
- (iii) all role eligibility, tensor, boundary, continuation, and failure constraints are preserved;
- (iv) no unused live target-relative residue remains;
- (v) the emission requires no selector, no carrier transfer, no post-hoc repair, and no silent scope reset.

This definition is independent of the theorem below: the theorem identifies when that standing-positive emission exists under certificate completeness.

Remark 3.19 (Terminology). The title has been strengthened to “Role-Occupancy Closure” because the operative theorem is stronger than cardinality matching. Role-cardinality closure remains as the finite cardinality shadow of occupancy closure.

Remark 3.20 (Constraint-diagnosis normal form: preview). Every attempted weakening of Role-Occupancy Closure must be located at one of five diagnostic loci or declared as explicit scope reset; no further same-regime route exists. The full proof is given in Section 8.

Diagnostic locus	Constraint-violation diagnosis
Candidate locus	a purported survivor is shown not actually admissible under C
Presentation quotient locus	a residue distinction is shown to be skin, notation, gauge, phase, coordinate, or bookkeeping only
Eligibility locus	a declared internal, provenance/indexical, or boundary-trace eligibility edge is exposed
Compatibility locus	a declared tensor, continuation, boundary, failure, or co-instantiation constraint is exposed
Role-mistyping locus	the target architecture is mistyped and must be reset explicitly; changing $\mathcal{R}_T$ is not same-certificate repair

4 Candidate classes do not realize themselves

Theorem 4.1 (Candidate-Class Non-Realization). *Let C be a fixed certificate and let $R_{\text{adm}}(C)$ be nonempty. The existence of admissible candidates does not by itself realize any member:*

$$R_{\text{adm}}(C) \neq \emptyset \not\Rightarrow \exists R_i \in R_{\text{adm}}(C) \text{ Realized}(R_i).$$

Proof. Candidate generation is not standing generation. Under the AASC kernel, standing cannot be created by internal production of a candidate list. Under standing conservation, standing can be preserved or destroyed but not generated, amplified, transferred, or repaired. Under no-generators, no internal operation that merely produces candidate classes confers realized status on one element of that class. Under no-carriers, a candidate does not inherit realized status by association with another successful or preferred candidate. Under no-repairs, later comparison cannot retroactively convert an unclosed candidate set into a realized same-act output. Therefore candidate status is necessary material for realization, but not realization itself. $\square$

5 Separation is not closure

Theorem 5.1 (Separation Is Not Role-Occupancy Closure). *Let $x, y \in \text{Residue}(C)$ with $x \neq y$. Their separation does not imply $\text{ROC}(C, T)$.*

Proof. The statement $x \neq y$ says only that two presentation-quotiented candidates are distinct. It does not determine whether either candidate is target-relative, which target roles they are eligible to occupy, whether all target roles are covered, whether a unique matching exists modulo certificate-lawful role-incidence automorphism, or whether the resulting assignment is jointly compatible. The candidates may be rivals for one role, complementary occupants of different roles, residues from different targets, residues outside $\text{Residue}_T^{\text{elig}}(C, T)$, pointwise eligible but not live under J_C , or mutually incompatible occupants. Therefore separation is not closure. $\square$

Example 5.2 (Three uses of separation). For neutrino ordering, two separated branches R_{NO} and R_{IO} are rivals for one ordering role. Separation yields two residues but two possible assignments to the same role, not a unique assignment. For a Dirac-style paired target, two separated sectors may

be complementary occupants of two distinct roles. For gauge or coordinate presentations, apparent separation may disappear before residue formation as presentation equivalence.

6 Forced role-cardinality and role typing

Theorem 6.1 (Forced Role Set). *Every realization target induces a forced role set:*

$$T(C) \Rightarrow \mathcal{R}_T(C, T).$$

The role set is fixed by target architecture and cannot be chosen after inspecting $\text{Residue}(C)$.

Proof. A construction cannot be realized until the role architecture it is supposed to populate is typed. Role typing fixes whether the target is singleton, paired, family-structured, multiplet-structured, quotient-structured, or sector-structured. If $\mathcal{R}_T(C, T)$ were chosen after inspecting $\text{Residue}(C)$, the target role set would not be forced by the target. It would be an after-the-fact selector. The AMetric boundary blocks such selection; fixed-domain exhaustion blocks undeclared role-set alteration; no-repair discipline blocks post-hoc target retagging. Therefore the role set must be target-induced before assignment. $\square$

Corollary 6.2 (No cardinality backfitting). *A certificate cannot license realization by redefining n_{role} to match the surviving residue after the fact. Such a move is selector import or scope reset, not same-scope correction to the closure diagnosis.*

7 Main theorem: unique compatible role occupancy

The biconditional form is split into necessity and sufficiency. The forward direction requires licensed realization. The reverse direction requires certificate completeness.

Definition 7.1 (Certificate completeness for role realization). A certificate C is role-realization complete for target T iff it is complete with respect to the following data:

- (i) candidate completeness: all candidate residues relevant to the target are included in $R_{\text{adm}}(C)$;
- (ii) quotient completeness: $\sim_{\text{pres}}$ collapses all and only presentation-level duplicates;
- (iii) role-set completeness: $\mathcal{R}_T(C, T)$ is target-forced and fixed before residue inspection;
- (iv) eligibility completeness: E_C records all same-certificate admissible target-role occupancies and only those generated by the formation rules for E_C ;
- (v) compatibility completeness: J_C records all certificate-level joint compatibility and failure conditions relevant to complete role occupation and only those generated by the formation rules for J_C ;
- (vi) live-residue completeness: every pointwise eligible but non-live residue has an explicit compatibility-failure or impossibility diagnosis in the certificate record;
- (vii) failure completeness: every impossible or non-live candidate has been rejected fail-closed rather than left as silent residue.

Theorem 7.2 (Necessity of compatible role-covering assignment). *If realization is licensed for (C, T) , then the set of compatible role-covering assignments has exactly one orbit under certificate-lawful role-incidence automorphism:*

$$\text{RealizationLicensed}(C, T) \quad \Rightarrow \quad \left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right| = 1.$$

Proof. If realization is licensed, every forced target role must have an admissible occupant. Otherwise the target is underpopulated and realization is incomplete. Every occupant must lie in $\text{Residue}_T^{\text{live}}(C, T)$: residues outside $\text{Residue}_T^{\text{elig}}(C, T)$ are irrelevant to target T , and residues in $\text{Residue}_T^{\text{elig}}(C, T) \setminus \text{Residue}_T^{\text{live}}(C, T)$ are pointwise eligible but jointly impossible. No live target-relative residue may remain unused, since such a residue appears in at least one jointly compatible complete occupation attempt for the target; retaining it unused would require a selector, a role split, or a declared scope reset. Thus realization induces a bijective role-covering assignment $\sigma : \mathcal{R}_T(C, T) \xrightarrow{\sim} \text{Residue}_T^{\text{live}}(C, T)$. Distinct forced roles receive distinct residues because role sharing has already been quotiented or retyped before the forced role set is fixed.

Pointwise eligibility follows because each assigned residue must be admissible for the role it occupies. Joint compatibility follows because realized co-occupation cannot violate tensor, boundary, continuation, or failure constraints. If two compatible assignments are distinct only by a certificate-lawful role-incidence automorphism, they are the same realization up to lawful relabeling of interchangeable elements of the role-incidence structure. If two assignments remain distinct after quotienting by $\Gamma_{C,T}^{\text{lawful}}$, the certificate leaves an admissibility-relevant choice between them. That choice would be a hidden same-scope selector, contrary to the AMetric boundary, no-carrier discipline, and fixed-domain exhaustion. Hence licensed realization entails exactly one assignment orbit. $\square$

Theorem 7.3 (Sufficiency under certificate completeness). *Assume C is role-realization complete for target T . If*

$$\left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right| = 1,$$

then realization is licensed for (C, T) .

Proof. A representative assignment $\sigma \in \Sigma_C(T)$ is a bijection: it covers every forced target role with an eligible and live target-relative residue, uses every live target-relative residue, assigns distinct residues to distinct roles, and satisfies the joint compatibility predicate. Candidate completeness ensures no relevant candidate is missing. Quotient completeness ensures no presentation duplicate is overcounted and no tensor-bearing distinction is wrongly collapsed. Role-set completeness ensures the target role architecture is not backfitted. Eligibility completeness and compatibility completeness ensure that no undeclared role-incidence or co-instantiation constraint remains outside the assignment. Failure completeness ensures that impossible candidates and pseudo-routes have already failed closed. The quotient by $\Gamma_{C,T}^{\text{lawful}}$ removes only lawful relabeling of the role-incidence structure; it does not introduce a selector. Therefore no selector is needed, no role is unpopulated, no excess live target-relative residue survives, and no joint incompatibility remains. By the independent definition of $\text{RealizationLicensed}(C, T)$, realization is licensed relative to the declared certificate. $\square$

Theorem 7.4 (Role-Occupancy Closure under Admissibility). *For a role-realization-complete certificate C , realization is licensed exactly when Role-Occupancy Closure holds:*

$$\boxed{\text{RealizationLicensed}(C, T) \quad \Longleftrightarrow \quad \left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right| = 1.}$$

Literal uniqueness of assignment is the special case $\Gamma_{C,T}^{\text{lawful}} = \{1\}$.

Proof. The forward direction is the necessity theorem. The reverse direction is the sufficiency theorem under certificate completeness.

Role-Occupancy Closure is a downstream tightening of the kernel invariants of admissibility, standing, reference, and irreversibility established in *Non-Degenerate Construction and the Kernel of Admissibility*. Its role is not to add a new generator of realization, but to express the finite target-occupation consequence of those invariants once a certificate and target are fixed. $\square$

Corollary 7.5 (Cardinality equality as consequence). *If $\mathcal{R}_T(C, T)$ and $\text{Residue}_T^{\text{live}}(C, T)$ are finite and $\text{RealizationLicensed}(C, T)$, then*

$$|\text{Residue}_T^{\text{live}}(C, T)| = |\mathcal{R}_T(C, T)| = n_{\text{role}}(C, T).$$

Proof. Every element of $\Sigma_C(T)$ is, by definition, a bijection $\mathcal{R}_T(C, T) \xrightarrow{\sim} \text{Residue}_T^{\text{live}}(C, T)$. The automorphism quotient identifies only equivalent bijections; it does not alter the domain or codomain of any representative assignment. Therefore the two finite sets have the same cardinality. $\square$

Corollary 7.6 (Singleton residue as special case). *If $|\mathcal{R}_T(C, T)| = 1$, then realization requires exactly one compatible assignment orbit to the single role. If $\text{Residue}_T^{\text{live}}(C, T)$ contains two eligible rivals for that role, closure fails even though both candidates may be individually admissible.*

Corollary 7.7 (Constraint-violation diagnostic table). *For finite live target-relative residue, failure of role-occupancy closure has the following basic dispositions:*

Condition	Constraint diagnosis	Meaning
$ \text{Residue}_T^{\text{live}} > \mathcal{R}_T $	over-residue	excess live target-relative residue survives
$ \text{Residue}_T^{\text{live}} < \mathcal{R}_T $	under-residue	some role cannot be populated
Equal cardinality but no compatible coverage	incidence failure	wrong residues for the roles
Incidence holds but $J_C = 0$	compatibility failure	assigned occupants cannot jointly realize
More than one orbit in $\Sigma_C(T)/\Gamma_{C,T}^{\text{lawful}}$	selector gap	multiple non-equivalent assignments survive
Exactly one orbit in $\Sigma_C(T)/\Gamma_{C,T}^{\text{lawful}}$	closure	realization licensed under complete certificate

8 Exhaustion of closure routes

Definition 8.1 (Impossible candidate). $\text{Impossible}_C(R_i)$ means that candidate R_i violates at least one non-negotiable component of C . A candidate is not impossible merely because it is unattractive, disfavored, unobserved, less probable, less elegant, or not selected.

Lemma 8.2 (Fixed-certificate diagnostic-locus normal form plus explicit reset). *Under a fixed certificate C , an alleged revision of the closure diagnosis must be located at one of the following diagnostic loci:*

- (i) candidate locus: a purported survivor is shown not actually admissible under C ;
- (ii) quotient locus: a purported distinction is shown to be presentation-only under $\sim_{\text{pres}}$;

- (iii) *eligibility locus: a same-certificate role-typing, provenance/indexical, or boundary-trace edge is exposed;*
- (iv) *compatibility locus: a certificate-declared tensor, continuation, boundary, failure, or co-instantiation constraint is exposed;*
- (v) *role-mistyping locus: the target role architecture is shown to have been mistyped, in which case correction requires explicit scope reset rather than same-certificate repair.*

Actual alteration of C , T , $\mathcal{R}_T$, admissibility typing, the boundary record, or the role vocabulary is not same-certificate closure. It is a reset $C \rightsquigarrow C'$ followed by fresh evaluation under the new certificate.

Proof. The assignment set $\Sigma_C(T)$ is determined extensionally by the live target-relative residue, the forced role set, eligibility, compatibility, and the certificate-lawful incidence action. The live target-relative residue is itself determined by $R_{\text{adm}}(C)$, $\sim_{\text{pres}}$, E_C , and J_C . Therefore, under fixed C , an alleged correction to the closure diagnosis cannot be an untyped alteration of those objects. It must be justified by a constraint diagnosis: a candidate was not actually admissible, a distinction was only presentation-level, an eligibility edge was already certificate-declared, or a compatibility condition was already certificate-declared. If the alleged change instead requires retyping the role set or changing the certificate, then the same-certificate problem has ended and a reset has begun. If no diagnostic locus is implicated, the same bijective assignments exist with the same compatibility values and the same $\Gamma_{C,T}^{\text{lawful}}$ -action; the closure diagnosis is unchanged. $\square$

Theorem 8.3 (Exhaustion of lawful closure routes). *A closure failure can be resolved only by one of the following lawful routes:*

- (i) *impossibility elimination: a purported survivor is shown to violate certificate components;*
- (ii) *presentation quotient collapse: a distinction is shown to be skin, notation, gauge, phase, coordinate, or bookkeeping only;*
- (iii) *eligibility exposure: an already-declared same-certificate eligibility edge is exposed;*
- (iv) *compatibility exposure: an already-declared tensor, continuation, boundary, failure, or co-instantiation condition is exposed;*
- (v) *explicit role reset: the target architecture is explicitly retyped under a new or reset certificate.*

All other attempted routes are selector import, empirical fit, residual matching, likelihood preference, post-hoc repair, diagnostic-only comparison, or bookkeeping.

Proof. By the fixed-certificate diagnostic-locus lemma, any same-certificate correction to the closure diagnosis must be located at the candidate, quotient, eligibility, or compatibility locus. The candidate locus is lawful only when a survivor is shown to violate certificate components; otherwise it is arbitrary exclusion. The quotient locus is lawful only for presentation-level duplicates; otherwise it erases tensor-bearing distinctions. The eligibility locus is lawful only when a same-certificate source of eligibility is exposed, such as internal role typing, provenance/indexical typing, or boundary trace. The compatibility locus is lawful only when a certificate-declared compatibility condition is exposed; otherwise it is post-hoc repair. The role-mistyping locus does not provide same-certificate alteration of $\mathcal{R}_T$; it requires explicit reset. Finally, changing C is explicit scope reset, not same-scope correction to the closure diagnosis. Since these diagnostic loci plus explicit reset exhaust the determinants of $\Sigma_C(T)/\Gamma_{C,T}^{\text{lawful}}$, no additional closure route remains. $\square$

9 Dirac-style paired solution diagnosis

The Dirac-style case is not used as a physical derivation. It is a schematic diagnosis for the singleton criterion boundary condition. If a formal system produces two residues and the target role architecture is one-role, the result is over-residual. If the target role architecture is paired-role and the two residues are eligible for complementary roles with joint compatibility, the result may be closed.

Target architecture	Residue situation	ROC status
one role	two eligible rivals	selector gap
two rigid complementary roles	unique compatible bijection	closed
two interchangeable role positions	one assignment orbit modulo automorphism	closed
two roles but both residues eligible only for one role	incidence failure	
two roles with incompatible occupants	compatibility failure	

The lesson is not that two solutions are always real. The lesson is that multiplicity must be interpreted through the forced role architecture, role eligibility, joint compatibility, and certificate-lawful role-incidence automorphism structure.

10 Neutrino ordering diagnosis

The neutrino ordering target in the AOT arc is a one-role target: one realized mass-ordering branch in one realized universe. The current residue contains two separated candidates, normal ordering and inverted ordering, under the current certificate.

Constraint coordinate	Neutrino-ordering status
$\mathcal{R}_T$	$\{r_{\text{ordering}}\}$
$\text{Residue}_T^{\text{live}}$	$\{R_{NO}, R_{IO}\}$
E_C	$E_C(R_{NO}, r_{\text{ordering}}) = 1, E_C(R_{IO}, r_{\text{ordering}}) = 1$ under the current certificate
J_C	each singleton assignment is internally compatible; no certified singleton trace or impossibility exclusion selects one
$\Gamma_{C,T}^{\text{lawful}}$	trivial for the one realized ordering role and its residue candidates
ROC status	two non-equivalent compatible assignments; no pre-lock prediction under current certificate

Proposition 10.1 (AOT neutrino residue is over-residual for a one-role target). *Under the current AOT certificate, the neutrino ordering diagnosis has*

$$\mathcal{R}_T(C_\nu, T_\nu) = \{r_{\text{ordering}}\}, \quad \text{Residue}_T^{\text{live}}(C_\nu, T_\nu) = \{R_{NO}, R_{IO}\}.$$

Both candidates are eligible for the same role and no singleton trace or impossibility exclusion has been certified. Therefore $|\Sigma_C(T)/\Gamma_{C,T}^{\text{lawful}}| = 2$, and $\text{RealizationLicensed}(C_\nu, T_\nu)$ fails.

Proof. The target asks for one realized ordering role. The current residue supplies two branch candidates, normal ordering and inverted ordering. Both are eligible rivals for the same role under the current certificate. The trace route is deferred because no singleton matter-response trace is certified. The exclusion route is deferred because neither branch is impossible under the current exclusion certificate. Therefore the certificate admits two assignments to the same role rather than one. By the main theorem, realization is over-residual and the selector is undefined. $\square$

Remark 10.2 (Closure routes for the ordering target). The status changes only when a certificate supplies a singleton trace, an impossibility object excluding one branch, or a forced role retyping under which the target no longer asks for one realized ordering branch. Absent such a route, empirical branch comparison remains post-lock comparison, not pre-lock realization.

11 Charged-lepton pole transport diagnosis

The charged-lepton readout arc separates three layers:

- (i) a carrier-level physical charged-lepton mass-shape residue;
- (ii) a symbolic role-projective readout and one-anchor metric import rule;
- (iii) an exact pole-skin transport target containing a protected finite trace remainder.

The pole target is not closed by reusing carrier closure alone. Exact pole masses require a further carrier-to-pole transport certificate.

Constraint coordinate	Charged-lepton protected-trace status
$\mathcal{R}_T$ Residue $_T^{\text{live}}$	$\{r_\ell^{\text{prot}}\}$, the protected pole-trace remainder role candidate protected trace remainders surviving carrier-to-pole transport diagnosis
E_C	a residue is eligible only if produced by declared pole-transport structure, not by empirical inversion or free counterterm choice
J_C	compatibility requires preservation of carrier residue, metric-anchor import, scheme discipline, and pole-readout constraints
$\Gamma_{C,T}^{\text{lawful}}$	trivial for the one protected remainder role and its live target-relative residues
ROC status	exact pole closure requires exactly one compatible singleton assignment; otherwise the pole target remains undetermined

Proposition 11.1 (Charged-lepton protected trace singleton requirement). *A charged-lepton carrier-level closure does not by itself license exact pole closure. Exact pole closure for the protected trace remainder requires a unique compatible role assignment for the pole-transport target. This diagnosis is criterion-setting rather than closed: the theorem states the exact closure condition the charged-lepton pole target must meet; it does not assert here that the current certificate already contains the unique protected-trace residue.*

Proof. The pole target is a distinct observable-transport presentation. The carrier-level residue may close symbolically and the one-anchor metric import may be conditionally licensed, but exact pole closure requires transport from the carrier object to pole-skin presentation. The protected trace remainder is a one-role target. Unless candidate protected trace remainders reduce to a single eligible and jointly compatible occupant of that role, the pole target remains underdetermined. Treating the residual as a free coefficient, finite counterterm, scheme conversion, target-leaking repair, or empirical inversion would be a pseudo-route. Therefore role-occupancy closure imposes the singleton-compatible-assignment condition. $\square$

12 Cosmological transport diagnosis

Cosmological parameter comparison is a same-target transport problem. A late-time value is not automatically the same parameter as an early-universe anchor merely because the same symbol is used. The transport certificate must preserve anchor data, tensor witnesses, quotient observables, continuation, boundary records, estimator identity, and hidden-remainder closure.

The cosmology transport paper defines a violation functional

$$V_{\text{AASC}}(R) = \sum_{j \in \{A, T, Q, C, B, E, M\}} \alpha_j C_j(R), \quad \alpha_j > 0,$$

whose zero set is the admissible transport solution space. The seven coordinates correspond to anchor data, tensor witnesses, quotient observables, continuation, boundary records, estimator/inference rule, and hidden same-scope remainder.

Constraint coordinate	Cosmological transport status
$\mathcal{R}_T$ Residue $_T^{\text{live}}$	$\{r_{\text{bridge}}\}$, one same-parameter early-to-late bridge role candidate transport-correction classes, estimator moves, boundary shifts, or hidden remainders eligible for the bridge role
E_C	eligibility requires declared transport typing rather than free fit improvement
J_C	compatibility requires all seven residual coordinates to vanish or lie within declared tolerance envelope
$\Gamma_{C,T}^{\text{lawful}}$	trivial for the one bridge role and its live target-relative residues
ROC status	the retained implementation does not supply a SH0ES-scale unique compatible same-target bridge under the tested transport basis

Proposition 12.1 (Transport residue is not realized bridge). *A same-target early/late cosmological comparison requires one realized transport certificate. Multiple candidate transport-correction classes do not realize the bridge unless there is a unique compatible assignment to the transport-bridge role and the joint compatibility predicate vanishes all transport residuals.*

Proof. The target is a single same-parameter bridge from early anchor to late-sector presentation. Thus the bridge target has one role. Candidate transport corrections, estimator adjustments, boundary-record shifts, hidden remainders, or quotient changes are candidate response classes. A role assignment is compatible only if the joint transport predicate closes all seven coordinates;

equivalently, in the numerical formulation, $V_{\text{AASC}}(R) = 0$ relative to the declared envelope. Outside that zero set, the residual manifests as anchor split, tensor omission, quotient mismatch, continuation patch, boundary repair, estimator selector, or hidden remainder, not as a realized same-target bridge. $\square$

In the retained numerical realization reported in the source transport paper, the constrained posterior remains close to the Planck/BAO regime and the present-epoch transport correction is statistically consistent with zero. The source explicitly reads this as a conditional non-resolution under the tested transport-closure constraints and low-dimensional basis, not as a rejection of SH0ES, not as a model-selection result against unrestricted Λ CDM, and not as an exclusion of all possible new-physics extensions. In ROC terms, the implemented transport basis does not supply a unique compatible same-target bridge to the SH0ES-scale value.

13 Yang–Mills sector-extension diagnosis

The Yang–Mills endpoint source separates a fixed local theorem object from extended-support sector structure. A bounded-region local gauge-invariant net and its vacuum-sector mass-gap theorem may close locally while extended-support sectors remain outside the bounded-region local algebras.

Constraint coordinate	Local target T_{local}
$\mathcal{R}_T$	roles needed to realize the bounded-region local gauge-invariant net and vacuum endpoint
$\text{Residue}_T^{\text{live}}$	local theorem-object residues
E_C	eligibility fixed by local net/vacuum theorem construction
J_C	compatibility fixed by local algebra, vacuum representation, and spectral endpoint
ROC status	local endpoint can close without populating extended-support sector roles
Constraint coordinate	Sector-extension target T_{sector}
$\mathcal{R}_T$	local theorem roles plus extended-support sector/realization roles over the fixed local net
$\text{Residue}_T^{\text{live}}$	sector residues attached to the theorem-scope extended-support class
E_C	eligibility requires sector realization over the fixed local theorem object, not bounded-local-algebra membership
J_C	compatibility requires preservation of the fixed local theorem object and no new subgap local-state content in same-theory completion
ROC status	recomputed relative to the enlarged target; local closure alone does not populate sector roles

Proposition 13.1 (Local closure does not populate extended-support roles). *A local Yang–Mills endpoint closure does not automatically populate extended-support sector roles. If the target is enlarged from the local theorem object to a sector-extension object, the forced role set must be recomputed.*

Proof. The local theorem object and the extended-support sector layer are different targets. The local net is a fixed base object; the sector layer is organized over it and carries global-form data invisible to strictly local observables. If the target remains local, sector roles are not part of the role architecture and need not be populated. If the target is enlarged to include extended-support sectors, the role set expands. Treating local closure as if it already populated the extended-support roles would be carrier transfer. Treating the sector layer as a same-scope modification of the local net would be theorem-object drift. ROC therefore requires recomputing role occupancy under the enlarged sector target. $\square$

Remark 13.2 (Clean diagnosis). This is the cleanest non-singleton diagnosis supplied by the theorem. The local target can close with one role architecture, while a sector-extension target has a different role architecture. Same-theory completion preserves the fixed local theorem object; sector-role population is an additional role-occupancy problem over that fixed base.

14 Standard objections and replies

14.1 Cardinality equality is insufficient

Correct. The theorem no longer uses bare cardinality equality as the main condition. Equal target-relative cardinality is a corollary of a unique compatible role-covering assignment modulo certificate-lawful role-incidence automorphism. Incidence and joint compatibility are explicit.

14.2 Why role-equivalence cannot be primitive

Correct. The manuscript uses presentation equivalence $\sim_{\text{pres}}$ for residue formation. Role occupancy is handled separately by E_C , J_C , and σ . The quotient no longer assumes the role assignment it is meant to help test.

14.3 Not every broad residue belongs to the target

Correct. The theorem uses the live target-relative residue $\text{Residue}_T^{\text{live}}(C, T)$, not the entire broad residue $\text{Residue}(C)$. A residue generated by the broad certificate but ineligible for every role of T is outside the target-relative assignment problem.

14.4 Role symmetries can produce many bijections

Correct. The theorem counts compatible assignments modulo $\Gamma_{C,T}^{\text{lawful}}$. Literal uniqueness is asserted only when the roles and live target-relative residues are rigidly individuated.

14.5 Compatibility is missing

No. The predicate $J_C(\sigma)$ is part of the main theorem and has formation rules. Pointwise role eligibility does not suffice unless the certificate proves joint factorization.

14.6 The biconditional is too strong

The biconditional is conditional on role-realization completeness. Necessity and sufficiency are proved separately. Without certificate completeness, only the necessity theorem is claimed.

14.7 The predicates EC and JC are arbitrary

No. Both are constrained by formation rules. Eligibility can arise under fixed C only from internal role typing, provenance/indexical typing, or a declared stable bivalent boundary trace. Scope reset creates a new relation $E_{C'}$ under a new certificate. Compatibility can arise only from declared tensor, continuation, boundary, failure, and co-instantiation constraints. Preference, success, scalar ranking, and repair are explicitly excluded.

14.8 This just says do not choose arbitrarily

No. The theorem distinguishes candidate generation, presentation quotienting, live target-relative residue, role eligibility, role coverage, certificate-lawful role-incidence automorphism, joint compatibility, and scope reset. It gives a typed assignment condition under a fixed target role architecture.

14.9 Dirac-style cases are schematic

Dirac-style cases show why singleton residue should not be treated as a universal rule. The theorem is therefore role-occupancy based rather than singleton based. The Dirac-style discussion is schematic only and is not used as historical or physical evidence.

14.10 Empirical data can pick the residue

Empirical comparison can identify an observed branch after the fact. It cannot retroactively license pre-lock realization under a fixed certificate. If empirical data are part of the declared certificate as a stable boundary trace or record, then the theorem uses that declared trace, not unrestricted post-hoc preference.

15 Relation to existing work

This paper operates inside the Admissibility And Standing Constraint framework and does not compete with generative frameworks in proof theory, model theory, category theory, or mainstream physics. Its aim is narrower: to tighten the residue condition by exhausting weaker realization criteria under the kernel invariants of admissibility, standing, reference, and irreversibility.

Nearby mathematical themes include normalization and proof-theoretic discipline, focusing in sequent calculi, definable-choice and elimination-style questions in model theory, uniqueness or lifting arguments in category-theoretic settings, and representation-theoretic sector organization in algebraic quantum field theory [16, 17, 18, 19, 20, 21, 22, 23, 24]. These neighboring frameworks typically presuppose an already-governed formal regime. The present paper instead works within AASC's fixed-domain constraint setting and asks when admissible residue can occupy a forced target-role architecture without selector import, repair, carrier transfer, or target drift.

The physical sections are therefore constraint diagnoses, not new generative models or empirical resolutions. The neutrino, charged-lepton, cosmological, and Yang–Mills discussions indicate how the role-occupancy condition locates a remaining constraint violation or target-typing distinction; they do not replace domain-specific dynamics, measurement, or sector construction.

The biconditional formulation and the fixed-certificate diagnostic-locus normal form are suitable for Lean4 formalization once the kernel package is fixed. The present manuscript treats that point as a formalization posture rather than as an additional premise.

16 Conclusion

Role-Occupancy Closure under Admissibility is the general finite-residue realization theorem. The singleton case is recovered when the forced target role set has exactly one occupant. The finite role-cardinality equation is recovered only as a cardinality shadow: equal numbers of residues and roles do not guarantee that the right residues occupy the right roles, that the residues are live under joint compatibility, or that the resulting assignment is unique up to only lawful relabeling.

The closure condition is:

$$\boxed{\text{RealizationLicensed}(C, T) \iff \left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right| = 1}$$

where

$$\Sigma_C(T) = \left\{ \sigma : \mathcal{R}_T(C, T) \xrightarrow{\sim} \text{Residue}_T^{\text{live}}(C, T) : \forall r \in \mathcal{R}_T(C, T) \ E_C(\sigma(r), r) \text{ and } J_C(\sigma) = 1 \right\}.$$

Thus every compatible assignment is a bijection from forced roles onto the live target-relative residue. The quotient by $\Gamma_{C,T}^{\text{lawful}}$ removes only certificate-lawful relabelings preserving standing status, tensor content, boundary records, continuation structure, failure conditions, and lawful redescription. Cardinality equality

$$|\text{Residue}_T^{\text{live}}(C, T)| = |\mathcal{R}_T(C, T)| = n_{\text{role}}(C, T)$$

is a necessary consequence, not the whole closure condition.

When no compatible assignment exists, the target is incomplete, incidence-incoherent, or compatibility-incoherent. When more than one assignment orbit exists, the certificate leaves a selector gap. When exactly one assignment orbit exists under a complete certificate, the target role architecture is populated without selection, repair, transfer, or theorem-object drift. Singleton residue is preserved as the $n_{\text{role}} = 1$ case; paired, family, multiplet, transport, and sector-extension targets are handled without distortion.

All uniqueness and non-existence claims in this paper are restricted to the fixed certificate C , target T , and governance-role equivalence on that domain. AASC supplies no generator for role occupancy. It constrains the live target-relative admissible residue until the forced role structure has a unique compatible bijective occupant assignment modulo only certificate-lawful equivalence. Any appearance of selection or generation is diagnosed as degeneracy: selector gap, incidence failure, compatibility failure, impossible residue, or explicit scope reset.

A Formal status ledger

Table 1: Main theorem stack after formal hardening.

Result	Function in this paper
Presentation equivalence	prevents circular role quotienting
Live target-relative residue	prevents broad non-target candidates from entering the role-covering theorem
Role eligibility formation rules	prevents arbitrary theorem-forcing eligibility edges
Joint compatibility formation rules	prevents arbitrary theorem-forcing compatibility predicates

Result	Function in this paper
Role automorphism quotient	handles interchangeable role positions without false non-uniqueness
Independent licensed-realization definition	prevents the main theorem from becoming merely definitional
Candidate-Class Non-Realization	blocks treating candidate sets as realized outputs
Separation Is Not Role-Occupancy Closure	blocks branch separation as selection
Forced Role Set	prevents cardinality backfitting after residue inspection
Necessity theorem	licensed realization implies one compatible assignment orbit
Sufficiency theorem	one compatible assignment orbit licenses realization under certificate completeness
Role-cardinality corollary	cardinality equality follows from occupancy closure
Fixed-certificate diagnostic-locus normal form plus reset	locates attempted closure at fixed-certificate diagnostic loci and explicit reset
Constraint diagnoses in one-role, paired-role, transport, and sector-extension targets	apply the constraint across paired-role, one-role, transport, and sector-extension target architectures

B Symbol ledger

Table 2: Symbol ledger.

Symbol	Meaning
C	fixed certificate
D	declared domain inside the certificate
T	target role architecture under evaluation
$R_{\text{adm}}(C)$	admissible candidate response set under C
$\sim_{\text{pres}}$	presentation-equivalence relation
$\text{Residue}(C)$	$R_{\text{adm}}(C)/\sim_{\text{pres}}$, the presentation-quotiented residue
$\text{Residue}_T^{\text{elig}}(C, T)$	eligible target-relative residue with at least one same-certificate eligibility edge for T
$\text{Residue}_T^{\text{live}}(C, T)$	live occupation-capable target-relative residue appearing in a jointly compatible pre-assignment
$\mathcal{R}_T(C, T)$	forced target role set
$n_{\text{role}}(C, T)$	$ \mathcal{R}_T(C, T) $
E_C	role-eligibility relation
$J_C(\sigma)$	joint compatibility predicate for a complete assignment
σ	bijective candidate role assignment from roles to live target-relative residues
$\Sigma_C(T)$	set of compatible bijective role-covering assignments

Symbol	Meaning
$\Gamma_{C,T}^{\text{lawful}}$	certificate-lawful role-incidence automorphism group preserving standing, tensor, boundary, continuation, failure, and redescription structure
$\text{RealizationLicensed}(C, T)$	realization licensed relative to certificate and target
R_{NO}, R_{IO}	normal-ordering and inverted-ordering branch residues
V_{AASC}	cosmological transport-closure violation functional

C Compact objection map

Table 3: Standard objections and replies after final formal hardening pass.

Objection	Blocking result
Candidate set realizes itself	Candidate-Class Non-Realization
Separated branches imply selection	Separation Is Not Role-Occupancy Closure
Every target must be singleton	Singleton is only $ \mathcal{R}_T = 1$
Multiple residues are always failure	Multi-role closure allowed by unique compatible assignment orbit
Bare cardinality equality suffices	Rejected; incidence and J_C are required
Role quotient is circular	Replaced by presentation quotient plus eligibility relation
Broad certificate residues must all be used	Replaced by eligible and live target-relative residue distinctions
Interchangeable roles or residues create false non-uniqueness	Assignments are quotiented by $\Gamma_{C,T}^{\text{lawful}}$, the certificate-lawful role-incidence automorphism group
Set n_{role} after the fact	Forced Role Set and no cardinality backfitting
Pointwise eligibility suffices	Joint compatibility predicate J_C required
E_C and J_C are ad hoc	Formation rules restrict both predicates
Use empirical fit as selector	Empirical material must be declared as trace or record
Use likelihood ranking	Ranking does not define unique compatible assignment
Add calibration after failure	Fresh evaluation or scope reset; no same-act repair
Quotient away inconvenient distinctions	Presentation quotient collapses only skin/presentation differences
Change certificate as same-scope patch	Certificate change is explicit scope reset
Local theorem closure populates sector roles	Recompute $\mathcal{R}_T$ when target architecture expands

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